SARA OMAR MAHMOUD

.NET Backend Developer

Cairo, Egypt | +20 106 730 4619 | saraomar12380@gmail.com
linkedin.com/in/sara-omar-9454b6115 | saraomar2016.github.io/sara-portfolio

PROFESSIONAL SUMMARY

.NET Backend Developer with 5+ years of experience building enterprise security, surveillance, and business systems, including license plate recognition, visitor and building management, command & control, access control, ticketing & tolling, and sales operations. Specialized in ASP.NET Core APIs, Clean Architecture, and database design with SQL Server and MongoDB. Experienced in real-time systems with SignalR and in Milestone XProtect integrations, including Smart Client plugins, live video, and playback. Delivered 10+ production systems from requirements analysis through deployment.

TECHNICAL SKILLS

- **Backend:** C#, .NET, ASP.NET Core, ASP.NET Web API, ASP.NET MVC, RESTful APIs, Entity Framework, LINQ, Swagger/OpenAPI
- **API Security & Reliability:** JWT Authentication, BCrypt Password Hashing, FluentValidation, Serilog
- **Architecture & Practices:** Clean Architecture, SOLID Principles, Dependency Injection, API Design, Database Design
- **Databases:** SQL Server (Stored Procedures, SqlTableDependency), MongoDB
- **Real-Time & Integrations:** SignalR, Milestone XProtect SDK (Smart Client Plugins, Live View, Playback), MailKit
- **Desktop & Cross-Platform:** WPF, Windows Forms, .NET MAUI
- **Reporting & Maps:** RDLC Reports, QuestPDF, EPPlus (Excel), LiveCharts, GMap.NET
- **Tools & Deployment:** Git, GitHub, Docker, Postman, Visual Studio, VS Code, Advanced Installer, Visual Studio Installer Projects

PROFESSIONAL EXPERIENCE

Freelance .NET Backend Developer May 2025 – Present

- Architected and delivered the **Mandoobi** backend end-to-end within a 4-person team: a retail and wholesale sales management platform covering sales representatives, supervisors, warehouses, vehicle inventory, invoices, quotations, money collection, and returns.
- Built an ASP.NET Core 9 Web API with 70+ controllers on MongoDB, structured with Clean Architecture and Dependency Injection.
- Secured the API with JWT authentication, BCrypt password hashing, and FluentValidation, and added Serilog structured logging.
- Implemented real-time updates with a SignalR hub, PDF quotations with QuestPDF, and email notifications with MailKit.
- Built 20 bilingual (Arabic/English) RDLC reports, including daily sales, inventory, profit margin, product performance, customer lifetime value, and salesperson performance.
- Implemented business logic and API integration for the WPF admin application on top of provided UI designs, and packaged it as a Windows installer.

.NET Backend Developer | MTI Holding — Cairo, Egypt March 2021 – May 2025

- As part of an 8-person development team, designed and built the backend for 10+ enterprise security, monitoring, and operations systems, including building management (BMS), visitor management (VMS), command & control center (CCC), entrance control, license plate recognition (LPR), and ticketing & tolling.
- Built **Milestone XProtect Smart Client plugins** for LPR and access control using the Milestone SDK, embedded live video streaming and playback into the Ticketing & Tolling System, and built video bookmarking and retrieval.

- Developed REST APIs with ASP.NET Web API and ASP.NET Core (Swagger) on MongoDB and SQL Server, shared by WPF desktop applications and a .NET MAUI tablet app.
- Delivered real-time features with SignalR and SQL Server change notifications (SqlTableDependency).
- Built GIS mapping tools for device placement using GMap.NET.
- Packaged and deployed the BMS, VMS, CCC, and entrance control desktop applications as Windows installers with Advanced Installer and Visual Studio Installer Projects.
- Designed SQL Server databases with Entity Framework and stored procedures for operational workflows.
- Built 20+ dynamic RDLC reports for the Ticketing & Tolling System (cashier shifts, refunds, subscriptions, auditing), plus LPR dashboards with charts and Excel export.
- Translated complex business requirements into database and application designs, and modernized legacy systems into a layered architecture.

KEY PROJECTS (MTI)

- **Security & Facility Management Platform:** Integrated BMS, VMS, CCC, entrance control, and time-limited camera access request modules for buildings, visitors, gates, and cameras. *Stack: ASP.NET Web API, ASP.NET Core, MongoDB, SignalR, WPF, .NET MAUI, RDLC*
- **LPR — Milestone XProtect Plugin:** Plate and car make recognition, black and white lists, live passing-car events, video playback, and analytics dashboards. *Stack: C#, WPF, Milestone SDK, MongoDB, SignalR, LiveCharts, EPPlus*
- **Ticketing & Tolling System:** Ticketing, tolling, monthly and UHF subscriptions, cashier shifts, 20+ dynamic reports, and embedded Milestone live video and playback. *Stack: C#, WinForms, SQL Server, Entity Framework, RDLC, Milestone SDK*
- **Company Management & Archive System:** Companies, agencies, contracts, products, policies, ratings, and maintenance notifications. *Stack: C#, WPF, SQL Server, Entity Framework, Excel export*

EDUCATION

Bachelor of Computer Science | Faculty of Computer & Information Sciences, Ain Shams University 2015 – 2020

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)